

Children's Medical Center of the Lakes

Department of Pathology, Pediatric Pathology Service
55 University Park Drive, Cedarlake, IL 60003 • CLIA 14D9933557

PEDIATRIC NEUROPATHOLOGY — POSTERIOR FOSSA RESECTION

Patient: WALKER, AIDEN T.	Accession: PED-26-N-00882
MRN: MRN-7720193	Collected: 2026-03-12
DOB / Age / Sex: 2017-11-18 / 8 y / Male	Reported: 2026-03-15
Ordering provider: Halpern, Nora B., MD (Pediatric Neurosurgery)	Location: Inpatient — Pediatric Neuro ICU

SPECIMEN

Cerebellar vermis mass, subtotal resection. Multiple soft, friable, gray-pink tissue fragments, aggregate 2.4 cm. No grossly identifiable necrosis. Submitted in cassettes B1-B7. Aliquots in RNAlater and snap-frozen for molecular and methylation profiling under separate accessions.

CLINICAL HISTORY

Previously well 8-year-old boy with three-week history of progressive morning headaches, ataxic gait, and one episode of emesis. MRI brain showed a 3.1 cm midline cerebellar vermis mass with obstructive hydrocephalus. No prior CNS history. No personal or family history consistent with Gorlin syndrome (no basal cell carcinomas, no rib or jaw anomalies, no palmar pits). Underwent posterior fossa craniotomy with subtotal resection on 2026-03-12. External ventricular drain placed peri-operatively.

INTRA-OPERATIVE CONSULTATION

Smear preparations of the posterior fossa mass demonstrate a densely cellular small round blue cell neoplasm with high nuclear-to-cytoplasmic ratio. Frozen section: "highly cellular small round blue cell tumor; differential includes medulloblastoma; defer specific subtyping to permanent sections, IHC, methylation, and molecular studies." Conveyed verbally to Dr. Halpern, 2026-03-12 11:18.

MICROSCOPIC DESCRIPTION

Sections demonstrate a densely cellular embryonal tumor composed of small round blue cells with hyperchromatic nuclei, scant cytoplasm, and brisk mitotic activity (greater than 15 mitoses per 10 high-power fields in hotspots). Homer Wright (neuroblastic) rosettes are identified focally. No large-cell or anaplastic features are appreciated and there is no significant nodularity or internodular pale-island architecture. Apoptotic bodies are scattered throughout. Adjacent cerebellar parenchyma shows reactive gliosis. No infiltrative pattern beyond the tumor mass is identified.

IMMUNOHISTOCHEMISTRY

Marker	Result	Comment
Synaptophysin	Diffuse, strong positive	Neuronal differentiation
NeuN	Focal positive	Neuronal lineage
INI1 (SMARCB1)	Retained nuclear expression	Excludes AT/RT
β-catenin	Cytoplasmic only (no nuclear translocation)	Argues AGAINST WNT-activated subgroup
GAB1	Diffuse cytoplasmic positive	Supports SHH subgroup at protein level
YAP1	Nuclear and cytoplasmic positive	Supports SHH or WNT (combined with β-catenin → SHH)
p53	Wild-type pattern (weak, patchy nuclear)	No aberrant overexpression
Ki-67 (MIB-1)	~45% in hotspots	High proliferation index, as expected for medulloblastoma
GFAP	Negative in tumor cells	Reactive astrocytes at margin only

DIAGNOSIS

POSTERIOR FOSSA / CEREBELLAR VERMIS, SUBTOTAL RESECTION:

— **MEDULLOBLASTOMA, CLASSIC HISTOLOGY.**

— **IHC pattern (GAB1+, YAP1+, β-catenin cytoplasmic, INI1 retained) favors SHH-activated subgroup.**

— **CNS WHO grade 4 (all medulloblastomas are grade 4 by definition under CNS5).**

— Final subgroup and risk stratification per integrated diagnosis once molecular and methylation classifier results return.

COMMENT

Morphology + IHC are diagnostic of medulloblastoma, classic histology pattern. Per WHO CNS5 (2021) and current pediatric neuro-oncology practice, integrated diagnosis requires assignment of one of four molecular subgroups: WNT-activated, SHH-activated (further stratified by TP53 status), or non-WNT/non-SHH (Group 3 or Group 4). The IHC profile here (GAB1+, YAP1+, β -catenin cytoplasmic) supports an SHH-activated subgroup; this is to be confirmed by methylation profiling and complemented by molecular sequencing for prognostically important variants (TP53, PTCH1, SUFU, SMO; MYC/MYCN amplification status). Integrated diagnosis to follow.

Electronically signed: Tan, Lin H., MD, Pediatric Neuropathology. Resident review: Doss, B., MD (PGY-4). Report status: FINAL (surgical pathology component). This component report is part of an integrated diagnostic episode.

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